Unconditional Proofs of Two Prime-Counting Assertions in OEIS A217864

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Abstract

The On-Line Encyclopedia of Integer Sequences (OEIS), entry A217864, records as conjectures that the number of primes in the intervals from $\lfloor n \log n \rfloor$ to $(n+1)\log(n+1)$ is unbounded, and that these intervals contain no primes infinitely often. We give concise unconditional proofs of both assertions. The first is a consequence of Maynard's theorem on bounded intervals containing arbitrarily many primes; the second follows from Westzynthius's theorem on arbitrarily large normalized gaps between consecutive primes. These are applications of established results, not claims of new prime-gap theorems.

1. Statement of the result

Throughout, log denotes the natural logarithm. For every integer $n \geq 1$, let

$$a(n) = \#\big\{p \text{ prime} : \ \lfloor n \log n \rfloor \leq p \leq (n+1)\log(n+1)\big\}. \tag{1}$$

This agrees with the definition of OEIS A217864 [1]. Put $b_n = \lfloor n \log n \rfloor$. Since p is an integer, the upper endpoint in (1) is equivalent to $p \leq b_{n+1}$; thus

$$a(n) = \#\{p \text{ prime} : b_n \leq p \leq b_{n+1}\}.$$

The use of closed intervals is intentional and matches the OEIS definition. Consecutive intervals can share an integer endpoint, but this does not affect either argument below.

Theorem 1. *The sequence $a(n)$ has both of the following properties, unconditionally:*

(i) $\limsup_{n\to\infty} a(n) = \infty$*;*

(ii) $a(n) = 0$ *for infinitely many positive integers n.*

2. Spacing of the endpoints

Let $d_n = b_{n+1} - b_n$. By the mean value theorem and the bounded effect of the floor function,

$$\begin{aligned} d_n &= (n+1)\log(n+1) - n\log n + O(1) \\ &= \log n + 1 + O(1). \end{aligned} \tag{2}$$

Moreover, $b_n \sim n \log n$. In particular,

$$d_n \longrightarrow \infty, \qquad \frac{d_n}{\log b_n} \longrightarrow 1 \quad (n \to \infty). \tag{3}$$

3. Proof of the theorem

Proof of Theorem 1(i). Fix an arbitrary integer $k \geq 1$. Maynard's bounded-prime-cluster theorem implies that there is a constant $H_k > 0$ for which arbitrarily large intervals of length H_k contain at least $2k$ distinct primes [2]. (Equivalently, apply his bounded-gap result to a cluster of $2k$ successive primes.)

Because $d_n \to \infty$, all sufficiently large consecutive endpoints b_n are separated by more than H_k. Consequently, any sufficiently far-out interval of length H_k meets at most two consecutive endpoint cells $[b_n, b_{n+1}]$. An interval containing $2k$ primes therefore has at least k of them in one of those two cells, by the pigeonhole principle. For that index n, the OEIS definition gives $a(n) \geq k$.

Maynard's prime clusters occur arbitrarily far out, so the resulting indices n can be taken arbitrarily large. Since k was arbitrary, $\limsup_{n\to\infty} a(n) = \infty$. □

Proof of Theorem 1(ii). Write $p_j < p_{j+1}$ for consecutive primes. Westzynthius's large-prime-gap theorem states that

$$\limsup_{j\to\infty} \frac{p_{j+1} - p_j}{\log p_j} = \infty; \tag{4}$$

see [3] or the stronger modern large-gap theorem in [4]. Hence there exist arbitrarily large consecutive primes $p < q$ with

$$q - p > 4 \log p. \tag{5}$$

For each such p, let n be the least index with $b_n > p$. Then $b_{n-1} \le p < b_n$. As $p \to \infty$, the endpoint asymptotics in (2)–(3) give

$$b_{n+1} - p \le (b_n - b_{n-1}) + (b_{n+1} - b_n) = (2 + o(1)) \log p.$$

In particular, for all sufficiently large chosen gaps, $p < b_n \le b_{n+1} < q$ by (5). There is no prime in $[b_n, b_{n+1}]$, since p and q are consecutive primes. Thus $a(n) = 0$. The indices n tend to infinity as $p \to \infty$, yielding infinitely many such zero terms. □

4. A general consequence of bounded prime clusters

Corollary 1. *Let $(x_n)_{n\ge1}$ be any strictly increasing, unbounded sequence of real numbers such that $x_{n+1} - x_n \to \infty$. Then the numbers of primes in the intervals $[x_n, x_{n+1})$ are unbounded.*

Proof. For each k, choose an interval of fixed length H_k containing $2k$ primes by Maynard's theorem. Sufficiently far out, it crosses at most one of the points x_n; thus at least k of its primes belong to one of two consecutive half-open cells. This can be done at arbitrarily large indices. □

5. Scope and attribution

OEIS A217864 labels both assertions in Theorem 1 as conjectures in its entry consulted on 25 October 2026 [1]. Its 2012 annotation already identifies the second assertion as following from the work of Brauer and Zeitz and describes a conditional route to the first. The present note records a direct, unconditional deduction of the first assertion using Maynard's 2015 theorem and supplies an explicit proof of the second using the classical large-gap theorem.

Novelty statement. Theorem 1 is a consequence of published theorems; no novelty for the prime-gap inputs, or priority for these deductions, is claimed. In particular, the result should be described as a clarification of the current status of two OEIS annotations rather than as the resolution of a previously open problem in the research literature.

References

[1] J. Perry and OEIS contributors, *The On-Line Encyclopedia of Integer Sequences*, A217864: "Number of prime numbers between floor($n \log n$) and $(n+1)\log(n+1)$," `https://oeis.org/A217864`. Accessed 25 October 2026.

[2] J. Maynard, "Small gaps between primes," *Annals of Mathematics* **181** (2015), no. 1, 383–413. `https://doi.org/10.4007/annals.2015.181.1.7`.

[3] E. Westzynthius, *Über die Verteilung der Zahlen, die zu den n ersten Primzahlen teilerfremd sind*, *Commentationes Physico-Mathematicae* **5** (1931), no. 25, 1–37.

[4] J. Maynard, "Large gaps between primes," *Annals of Mathematics* **183** (2016), no. 3, 915–933. `https://doi.org/10.4007/annals.2016.183.3.3`.

AI Authorship and Verification Disclosure

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